

ANEXO — Figuras

A. TTL (74LS04)

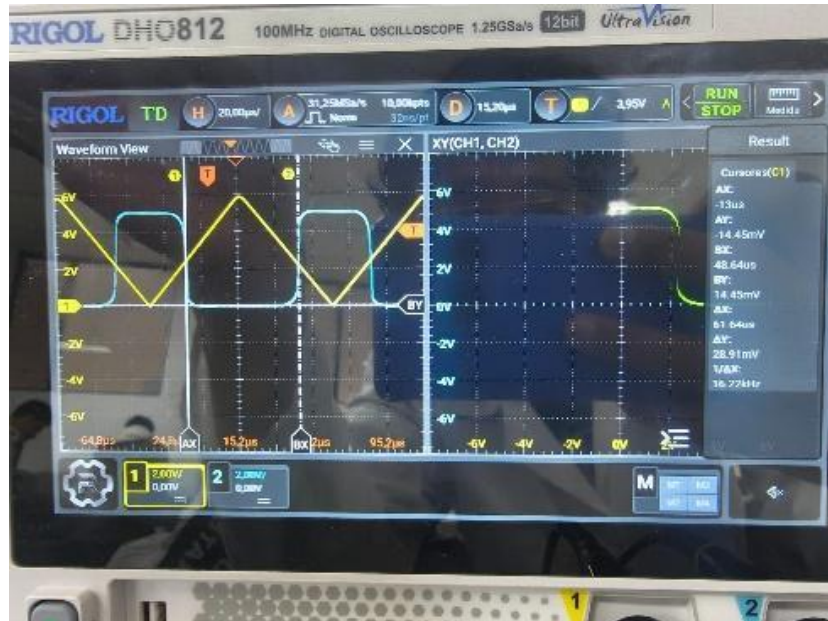


Figura 1. Función de transferencia V_{out} vs V_{in} del 74LS04 (modo XY; 4 Vpp, 1 kHz, offset 2 V).



Figura 2. Niveles lógicos del 74LS04 con cursores ($V_{OH} \approx 4.92$ V; $V_{OL} \approx 0$ V).



Figura 3. Tiempo de bajada del 74LS04: 24.0 ns.



Figura 4. Tiempo de subida del 74LS04: 41.0 ns.

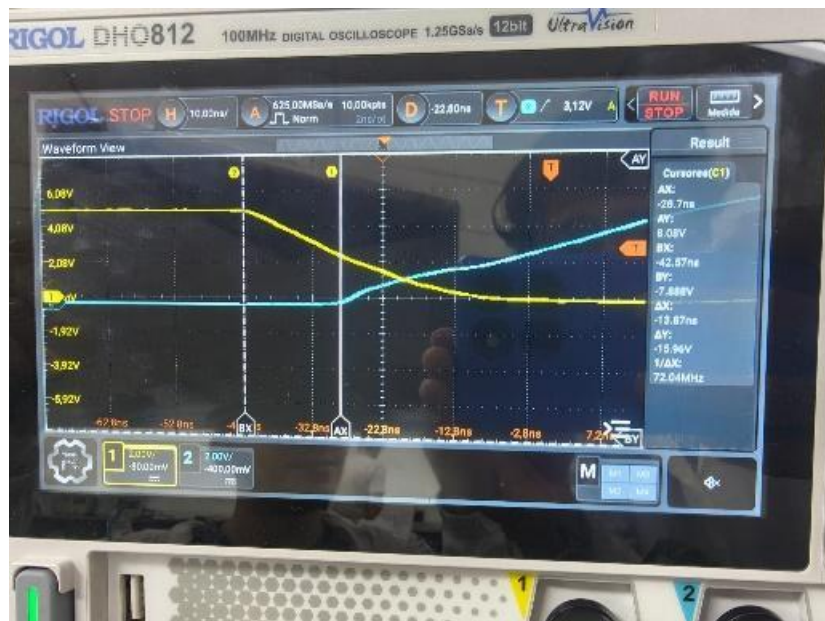


Figura 5. Retardo de propagación del 74LS04 (bajada): ≈ 13.9 ns.

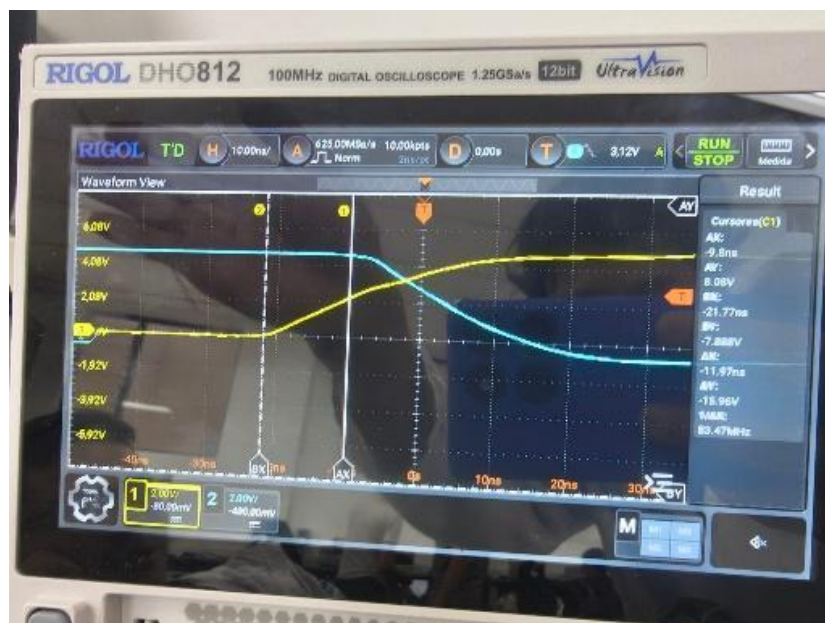


Figura 6. Retardo de propagación del 74LS04 (subida): ≈ 12.0 ns.



Figura 7. 74LS04 a 1 kHz.



Figura 8. 74LS04 a 10 kHz.



Figura 9. 74LS04 a 100 kHz.



Figura 10. 74LS04 a 500 kHz.



Figura 11. 74LS04 a 1 MHz.

B. CMOS (CD4069)

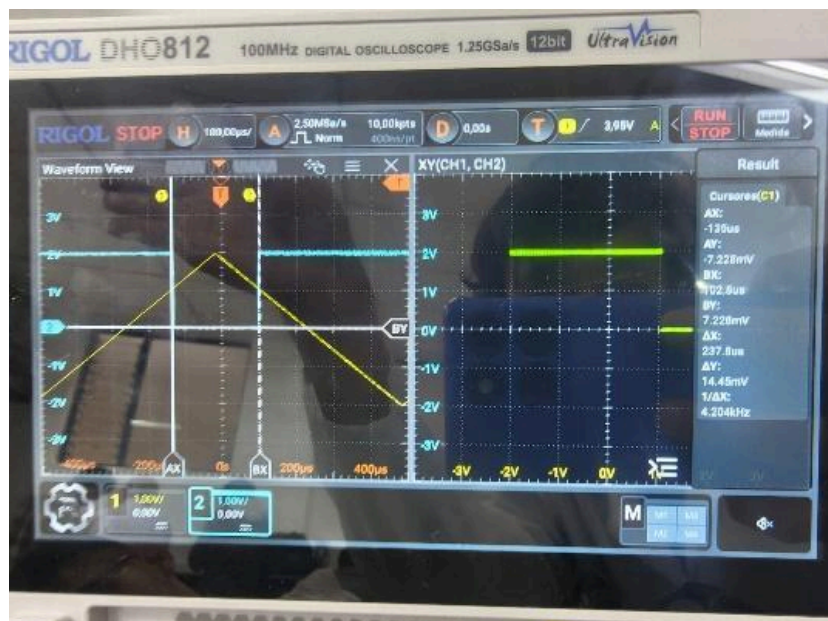


Figura 12. Función de transferencia V_{out} vs V_{in} del CD4069 (6 Vpp, 1 kHz, offset 3 V).



Figura 13. CD4069 a 1 kHz con cursores: $VOH \approx 4.95\text{ V}$ y $VOL \approx 11.97\text{ mV}$.



Figura 14. CD4069 a 10 kHz.



Figura 15. CD4069 a 100 kHz.



Figura 16. CD4069 a 500 kHz.



Figura 17. CD4069 a 1 MHz.

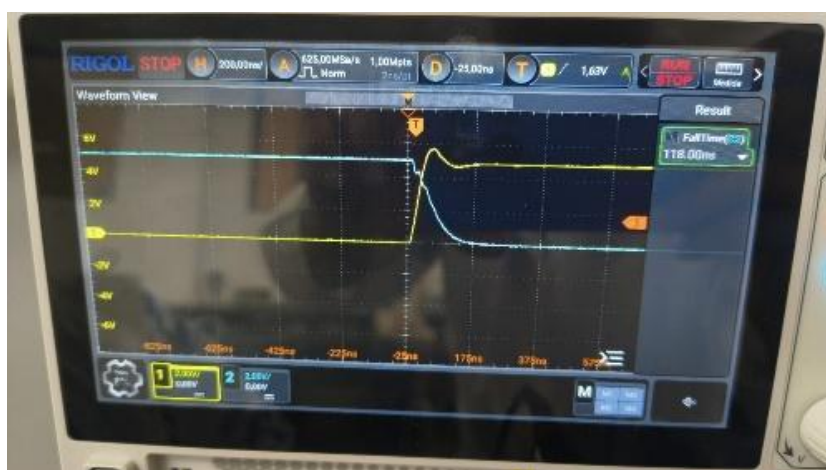


Figura 18. Tiempo de bajada del CD4069: 118 ns.



Figura 19. Tiempo de subida del CD4069: 160 ns.



Figura 20. Retardo de propagación del CD4069 (subida): ≈ 7.4 ns.

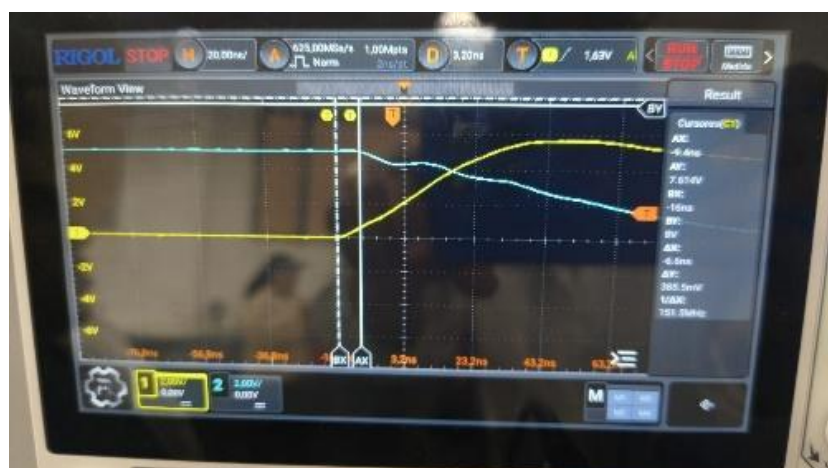


Figura 21. Retardo de propagación del CD4069 (bajada): ≈ 6.6 ns.

C. Compuertas AND / OR

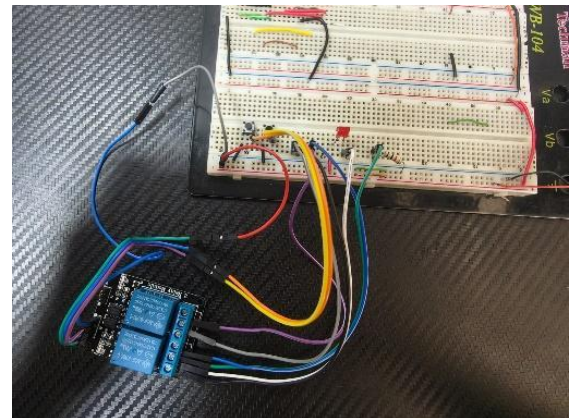
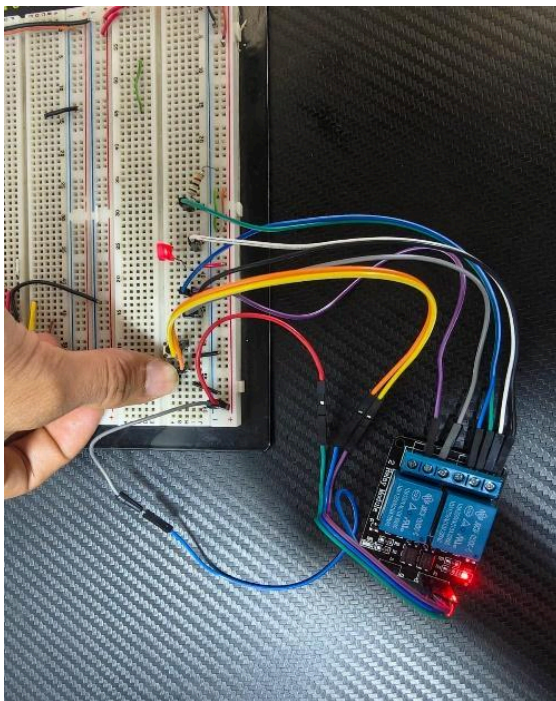
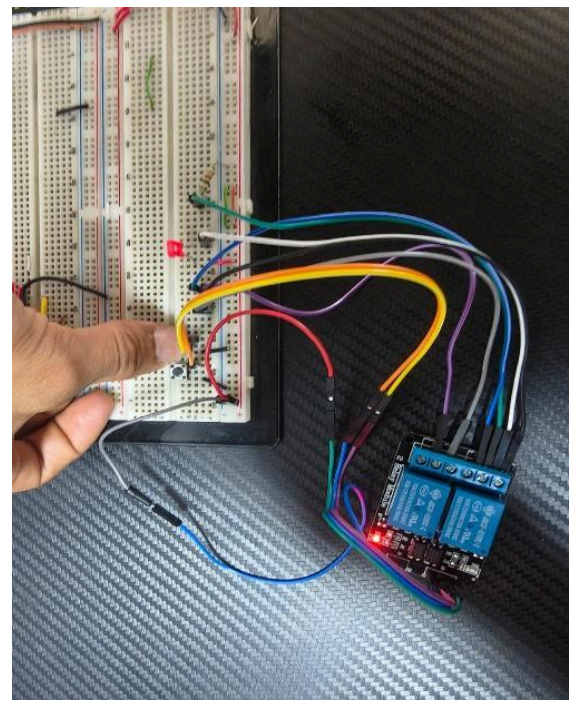
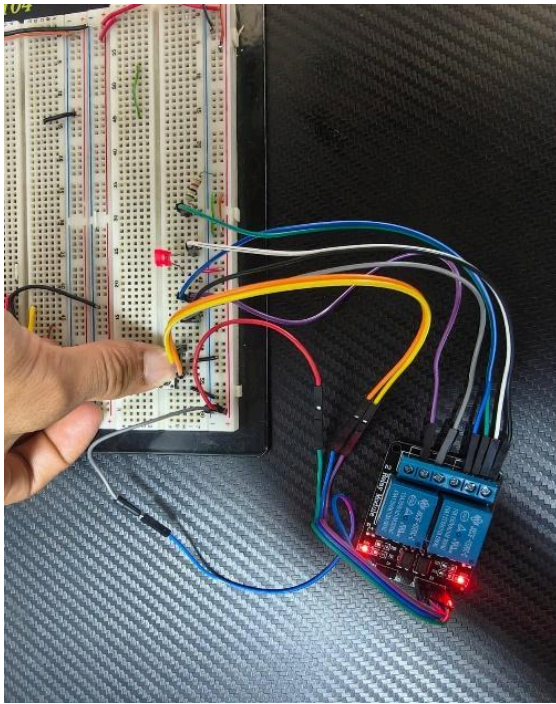


Figura 22. Compuerta OR con pulsadores y relés: el LED de la protoboard enciende con un relé o con ambos (secuencia: pulsador inferior, superior, ambos).

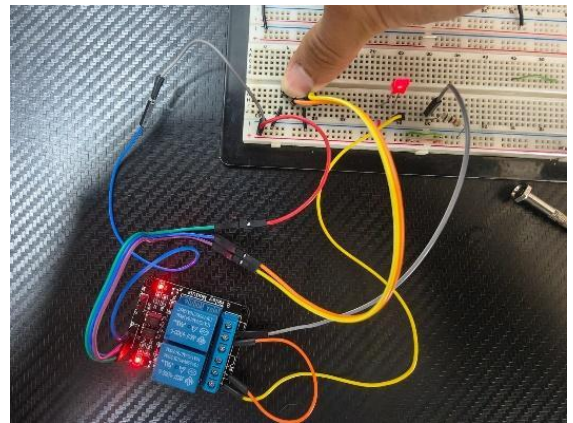
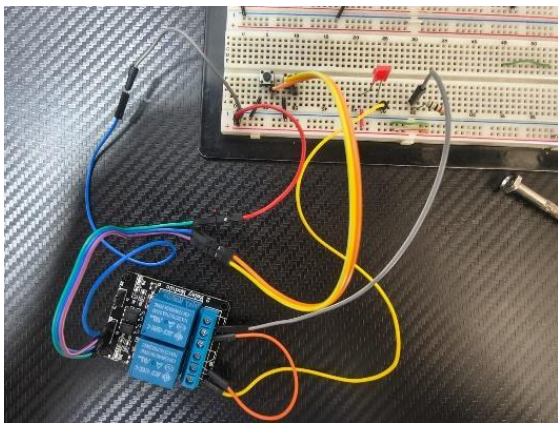
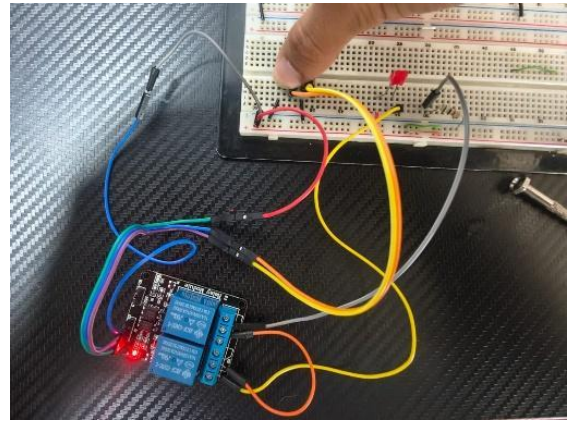
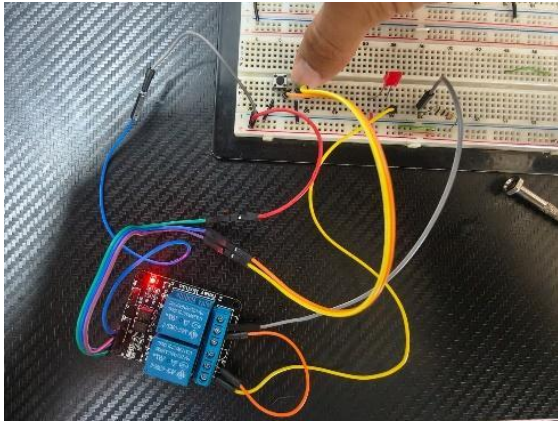


Figura 23. Compuerta AND con pulsadores y relés: el LED de la protoboard enciende solo con ambos relés activos.



Figura 24. Multímetro: (a) VOH TTL 2.50 V, (b) VOL TTL 21 mV, (c) VOH CMOS 1.88 V, (d) VOL CMOS 4 mV.

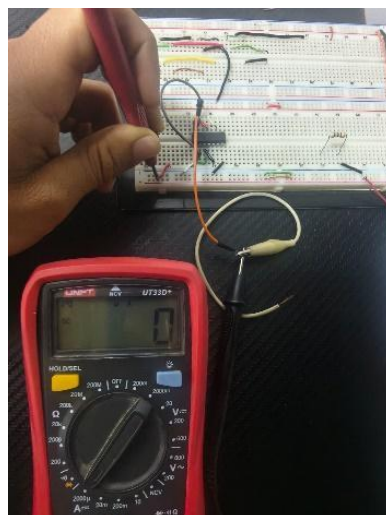
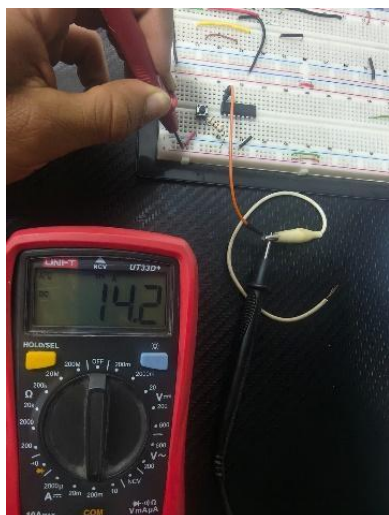


Figura 25. Corriente de alimentación: (a) ICC del TTL = 14.2 mA, (b) ICC del CMOS = 0 μ A.